The New Study Partner: How AI Is Changing University Learning in Kenya

With the emergence of generative AI into the daily lives of students and lecturers, Kenyan students and lecturers are facing a tough question: Does AI assist the students in learning or make them look good?

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Article

One night, about 11 p.m., Kimutai Samuel was not reading a textbook for an answer.

The 23-year-old Computer Science student at the Co-operative University of Kenya (CUK) was trying to get a microservices-based e-commerce backend running. His Spring Boot project was failing at the API Gateway, and the logs were returning an inexplicable “401 Unauthorized” error, despite his feeling the authentication token was valid.

He duplicated his repository settings and error log in Claude.

The solution is a simple one, he said: the security filters had to be ordered. The AI provided some refactored Java code, and the system started running.

This was an unusual thing for Samuel. It was just another late-night bug hunt with a tool they have become accustomed to using in their studies.

“That's simple, so your retention muscle goes soft," he replied.

In the question-and-answer session, that was the one that struck a nerve with universities in an era when AI is increasingly haunting them with the question: If AI can make a student faster, more productive, more able to find an answer, is the student learning more?

The question, which was once a mere speculation in Kenyan universities, is now gaining traction among students. With generative AI, you can get explanations of hard concepts, summarize papers, even write computer code, make outlines, and get answers anytime. Meanwhile, educators are struggling with academic integrity, unreliable information generated by AI, and the challenge of knowing if a student has copied and pasted what AI generated.

A study of 137 academics from 13 Kenyan universities in 2025 revealed that teachers were aware of the potential benefits of generative AI in the education sector, in terms of enhancing teaching, research productivity, and learner engagement. However, the same teachers voiced concerns about the ethical implications, academic integrity, and potential loss of valuable human aspects in learning. From the researchers' conclusions, it can be said that ethical policies, training, and institutional support would be needed for successful adoption.

The change is already evident at CUK on the campus.

Students are seated in a group following lectures, using laptop computers and phones. Assignments are discussed in groups by others. In classrooms, lecturers are still there, and students are getting a second explanation in their pockets.

There has been no replacement for the classroom with technology. Rather, it has added an extra player to it.

The student who requests AI to solve the issue

Samuel is known as a "prolific" user of AI.

He is using AI to debug, boilerplate code, refactor, and script. He doesn't see why he should take 30 minutes to write a ton of repetitive code to feed into a test database when he'll be creating hundreds of user profiles for a fictional country, in this case, Kenya.

He requests the AI to do it.

There is a difference between allowing AI to create something and grasping the content it produces, however, Samuel argues.

For example, in his late-night Spring Boot problem, he took the solution, as he needed his application to run. Next, he asked the AI to tell him why the security configuration was successful.

I certainly couldn't just leave it," he said. If it breaks during the defense, I'm toast!

His implementation of AI is not a mere replacement of work. It's all about minimizing time stuck.

He says that as much as 40% of the lines of code for some of his programming projects could be boilerplate or functions written by AI, but that the underlying structure would be his.

That's a tough one because as the machine does more of the programming, what is still considered to be the student's work?

“What's happening in the software industry is being reflected in the technology,” says Samuel. He also fears that there are students in his class who use AI without comprehending what it generates.

He has seen students create applications, make superficial modifications, and submit without being able to explain the basic parts of the code.

“You have to know how to code without it, if you don't, you're not a software engineer, you're a prompt operator.”

AI is not just a tool for productivity for him – it can be a trap.

It makes me feel better as an engineer but not so good as a student,” he said.

What he calls AI is very telling: “A jet engine mounted on a bicycle.”

But if the student knows how to steer, it can be extraordinarily fast, he explained.

The person who is teaching. The teacher is on the telephone.

20-year-old Shanice Shangi has a different perspective on AI.

The student at CUK using the Information Technology is not the main user for creating completed assignments. She employs it as a teacher.

In a Business Information Systems class, she had problems with distinguishing between two-tier and three-tier Enterprise Resource Planning (ERP) architectures. After studying in her hostel, she opened ChatGPT to get an explanation of the architecture from the system using a supermarket like Naivas to illustrate.

The important factor was not just getting a definition. She desired that the technology clarify where the database is, where the application logic runs, and what the cashier sees.

The explanation worked.

Then the diagram I had just copied into my lecture notes came to life, she said. I really did get it.

That distinction matters.

The ability to personalize learning is one of the most alluring educational applications of AI, as demonstrated by Shanice's use of the technology. If a student is not clear about an explanation given in class, they can ask for another explanation, and then another, without worrying that the person who is explaining will get impatient.

“It's a teacher who never gives up,” she said.

She also leverages AI to create summaries of academic papers and aids in idea organization during writer's block. In one essay, she leveraged AI to create an outline and subheadings and then wrote the paragraphs by herself.

But Shanice doesn't feel completely at ease with the impact AI has on learning.

She is concerned that students are not challenged intellectually enough.

“If you simply ask AI to write out the exam and pass it, and graduate from the school, what happens when a system crashes at the first job and there is no ChatGPT to tell you how to fix it step by step?” she asked.

Her worry is not isolated; it is a larger question that is growing in the field of AI: “Does enhanced performance imply enhanced learning?”

A 2025 meta-analysis that looked at 57 studies and 97 estimates of the impact of generative AI on learning revealed evidence of positive impacts for a range of learning outcomes, and a lack of a statistically significant impact on metacognition or monitoring and regulating one's own learning. The distinction is important because a student may be able to get the right solution without necessarily improving their ability to evaluate their own understanding of the problem.

The question is not just about the accuracy of AI answers.

It is how the student's thoughts happen to change as he receives them.

The student who left the class.

On the other side of the coin, Lokula Bob, 21, a Cyber Security student at the University of Nairobi, is the other student.

His experience with AI made him less reliant and more skeptical.

In one of his previous semesters, he was doing a reverse engineering laboratory task with malware, and he was working in a controlled sandbox. He was having trouble figuring out a series of Assembly instructions, and finally entered the instructions into ChatGPT.

A reply was received that passed the test.

It was also wrong.

The AI failed to understand part of the CF as it did not have the full memory-register context, Bob said. He shut the tool, went back to his book, and did the registers by hand.

He had concluded that a good answer does not always equate to a true one.

He told his lecturer he wished that he could show his real intellectual capacity if he were to get a B-, "takes a B-, not a machine's polished A.

Bob realizes that he is doing it slower.

Classmates can complete an assignment swiftly with AI support, but he has to spend more time reading technical documentation and do his work manually. The added work creates “permanent muscle memory,” he says.

He does not fear that students will be using AI!

It's that they will cease to develop their work without it.

He says that it's an "illusion of competence," the sense that you know something because a machine has explained it intelligently.

The illusion is broken if your lecturer asks you to explain your paper on the spot without your device," he said.

Bob's point of view is opposite to the one I just mentioned. While AI can help eliminate friction from learning, there are types of friction that can be part of learning.

Looking for an explanation, trying an idea, making a mistake, and trying again can be frustrating. They can also help to make one student independent.

When the lecturer observes

When Ian Leboo, a lecturer at CUK, noticed a gap between the written work by some students and their explanation of the work, AI had to be part of the picture.

He remembered one student who had a difficult time articulating in technical terms, and they took an assessment to design a system, and he marked it and was like, "Wow, he actually did a really good job at putting together an assessment for a system design. The papers submitted were particularly polished.

The tipping point for Leboo was when the student explained a concept from the paper without using it. The student was unable to describe what he had written, said Leboo.

That's the true challenge for the lecturer.

A student can't have just produced something with a machine. The bigger issue is that the student could earn a credit for something they're unable to prove.

Instead of just banning AI, Leboo feels there is a better way.

He likens the ban of the technology to banning the Internet or calculators. Rather, he believes the assessment of universities should change.

He is now more interested in oral defenses, in involving students in explaining the justification of their submission in a practical task.

“Don't try to trap the AI,” he said. “Test and test differently.”

That's not where he's going to be, as close to Samuel as it sounds.

He wants universities to be teaching the proper use of AI, such as auditing AI-generated code for security issues.

Leboo also feels that students should be taught to leverage AI; however, not to the detriment of their foundational thinking skills.

The difference lies in their limits.

AI is to Samuel like a tool to help a skillful student.

Leboo is concerned that students would use the tool even before they can assess its output.

The University was between the tool and the rule.

This can be seen in CUK's public academic spaces.

The University has put in place academic-integrity and assessment policies, and its library has access to Turnitin and links to a variety of AI research and productivity tools. CUK has also stated in public communications that academic integrity and examination standards are the key values that will be strengthened.

The extent to which CUK has a single, standalone institutional policy focused specifically on generative AI is not made public.

That distinction matters.

To term individual lecturers' practices as the university-wide AI policy would be misleading. The reaction to AI, on the other hand, is beginning to take shape in the current academic integrity and assessment frameworks, as well as through the individual department and lecturer's adaptation of their teaching.

It is reflective of a broader problem in Kenya.

It has been predicted that artificial intelligence and other new technologies will change the university landscape. In Kenya, researchers in higher education are also actively looking at the integration of AI in higher education institutions without compromising on issues of ethics and academic integrity.

According to a 2025 survey of university teachers in Kenya, perceived usefulness was the most important factor that made Kenyan university teachers want to use generative AI. Teachers were aware of its potential to help improve efficiency, facilitate literature searches, summarize academic content, and give individualized help, but were aware of concerns of integrity and the human aspect of education.

The evidence coming in from around the world is similar.

A study of university students has shown that students' competence with AI influences their perception and utilization of various generative-AI tools. Studies on students' perceptions of AI have revealed that students find it beneficial for various tasks, such as the preparation of assignments, but also warned of potential negative effects of its misuse and overuse, which could sabotage educational objectives.

The discussion is thus becoming more and more a matter of discussion.

The issue now is not just if students will use AI tools, but how. Now it's not just about whether students will use AI tools, but how.

They already are.

Whether universities will teach their students to use it critically is another question.

Catching up with AI to demonstrating comprehension

The impact may be a shift towards fundamental changes in assessment.

The traditional university assignment is based on the premise that the work is a reasonable representation of what a student knows and is able to do.

Generative AI undermines the assumption.

A well-formulated essay is now possible within seconds. A description of the desired application can be used to generate computer code. You can give a student with a problem understanding a concept an explanation that is customized to their specific question almost immediately.

That's not to say that the work is garbage.

It makes the finished product less revealing, however.

For the purposes of ascertaining a student's grasp of a subject, universities might have to look at the process as well as the product.

This can involve oral defenses, supervised practical examinations, demonstrations, project discussions and drafts, and other assessment methods where students have to justify decisions.

Leboo's strategy is along this line.

So does Bob's preferred means of demonstrating knowledge, instead of just presenting polished work.

Even the enthusiastic AI user, who is Samuel, knows that students can't do anything without AI.

The irony is that AI is likely to make universities better at measuring learning.

The final essay, examination script, or software project has long been used as an indicator of understanding for years. It's harder to make that assumption with AI.

The machine has not completely killed assessment.

It has revealed its faults.

The new study partner

The students resume using their phones and laptops on campus.

Some are a little more aggressive with AI. Some have selective uses. Some avoid it.

It is worth emphasizing their differences as an indicator that there is no homogeneous “student experience” of AI.

Samuel is in a hurry to get into action quickly.

Shanice wants an explanation that she can really understand.

Bob wants to retain the struggle, which he feels is his own knowledge.

Leboo wants students to demonstrate their understanding of their submissions.

Not one of them is saying that technology will be removed from the University.

The harder question is how to deal with it every day.

This could mean universities have to educate people on more than just how to operate an AI chatbot. Students must understand how to challenge the answers, check their sources, guard confidential information, spot bias, and understand that the most correct answer may not be the most instructive.

They also must be aware of what is still uniquely human.

Since Samuel is a member of a group project, a problem he encounters is the disagreements with his team members regarding the code and the late-night talk in the computer labs.

Shanice's pressure is to defend a project in front of an ACTIVE panel.

Bob sees it as the mentor who can see a student's potential and push them to do better.

For Leboo, the human dimension is mentorship and peer interaction.

Education is not about data transfer, it is emotional and social transformation,” said.

That's perhaps the key difference in the whole argument.

Artificial intelligence can produce information. It can clarify, summarise, generate, and correct. It can accelerate an individual student's progress significantly.

However, a university isn't just a place that passes information from one person to another.

It is a venue for students to make their own decisions, to be independent, to become confident, disciplined, and to argue for what they know.

Those responsibilities will not be taken away when AI comes into play.

It makes them more difficult to measure.

Maybe this is why Leboo's closing comment about AI is apt.

AI in the University's teaching is like a mirror for me, he said.

This technology is highlighting both strengths and weaknesses of the education system that has embraced it.

It lets students see the potential of a great new instrument.

It also demonstrates how a great end product can mask a lack of understanding on the part of the educator.

Thus, the debate between classrooms and artificial intelligence, or between traditional learning methods and technological shortcuts, may not be the future of university education in Kenya.

It can be a matter of learning how to operate the machine, rather than the machine learning how to operate you.

The jet engine is already ticking for students such as Samuel.

The next challenge is ensuring that they still know how to steer.